PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 2 · Version A · Solutions

21 questions · 90 minutes · original practice exam, not an official exam

ANSWER KEY

1	**c**	2	**d**	3	**c**	4	**a**	5	**d**	6	**b**	7	**b**
8	**a**	9	**d**	10	**a**	11	**c**	12	**e**	13	**b**	14	**e**
15	**a**	16	**c**	17	**a**	18	**c**	19	**c**	20	**a**	21	**b**

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 5 kg crate starts from rest and is pushed 8 m across a level floor by a 40 N force directed 20° below the horizontal. The coefficient of kinetic friction is 0.3.

1) How much work does the pushing force do on the crate?

Answer: (c) 301 J

$W = Fd\cos20° = (40)(8)(0.940) = 301$ J.

2) What is the speed of the crate after it has moved 8 m?

Answer: (d) 7.75 m/s

$N = mg + F\sin20° = 62.7$ N, friction does −151 J. $W_{net} = 150$ J $= \frac{1}{2}mv^2$, $v = 7.75$ m/s.

3) How much work does the normal force do on the crate?

Answer: (c) Zero

The normal force is perpendicular to the displacement.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A toy launcher has a spring with $k = 800$ N/m. It is compressed 0.15 m and fires a 0.2 kg ball straight up. Treat the launch point as the release point and ignore air resistance.

4) What is the launch speed of the ball?

Answer: (a) 9.49 m/s

$\frac{1}{2}kx^2 = 9$ J $= \frac{1}{2}mv^2$, $v = 9.49$ m/s.

5) How high above the launch point does the ball rise?

Answer: (d) 4.59 m

$mgh = \frac{1}{2}kx^2$, $h = 9/(0.2 \times 9.81) = 4.59$ m.

6) If the spring were compressed twice as far, the maximum height would be multiplied by

Answer: (b) 4

Spring energy $\propto x^2$, so the height is 4 times larger.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 2 kg ball on a 1.5 m string is pulled aside until the string makes 60° with the vertical, then released from rest.

7) What is the speed of the ball at the lowest point?

Answer: (b) 3.84 m/s

$h = L(1 - \cos 60°) = 0.75$ m; $v = \sqrt{(2gh)} = 3.84$ m/s.

8) What is the tension in the string at the lowest point?

Answer: (a) 39.2 N

$T = mg + mv^2/L = 19.6 + 19.6 = 39.2$ N.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1200 kg car accelerates from rest to 25 m/s in 9 s on a level road. Ignore friction and air resistance.

9) What average power does the engine deliver?

Answer: (d) 41.7 kW

$P = \Delta K/t = \frac{1}{2}(1200)(25^2)/9 = 41.7$ kW.

10) To reach 50 m/s from rest in the same time, the average power would need to be multiplied by

Answer: (a) 4

$K \propto v^2$, so doubling the final speed takes 4 times the energy in the same time.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 1500 kg car moving east at 20 m/s rear-ends a 2500 kg truck at rest. The two vehicles lock together.

11) What is their speed just after the collision?

Answer: (c) 7.5 m/s

$m_1v_1 = (m_1 + m_2)v$, $v = 30\,000/4000 = 7.5$ m/s.

12) How much kinetic energy is lost in the collision?

Answer: (e) 188 kJ

K before = 300 kJ, after = $\frac{1}{2}(4000)(7.5^2) = 112.5$ kJ, lost 188 kJ.

13) During the collision, the force of the truck on the car is

Answer: (b) equal in size to the force of the car on the truck.

Newton's third law: the forces are equal and opposite.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.5 kg ball moving at 4 m/s makes a head-on elastic collision with a 1.5 kg ball at rest.

14) What is the velocity of the 0.5 kg ball after the collision?

Answer: (e) 2.0 m/s backward

$v_1' = (m_1 - m_2)/(m_1 + m_2) \cdot v = (-1/2)(4) = -2$ m/s, so 2.0 m/s backward.

15) What is the speed of the 1.5 kg ball after the collision?

Answer: (a) 2 m/s

$v_2' = 2m_1/(m_1 + m_2) \cdot v = (0.5)(4) = 2$ m/s.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 0.15 kg baseball arrives at a bat at 35 m/s and leaves in the opposite direction at 45 m/s. The ball is in contact with the bat for 1.2 ms.

16) What is the magnitude of the average force of the bat on the ball?

Answer: (c) 1.00×10^4 N

$\Delta p = 0.15(35 + 45) = 12$ kg·m/s; $F = \Delta p/\Delta t = 1.00 \times 10^4$ N.

17) The impulse on the ball points

Answer: (a) in the direction of the ball's final velocity.

$\Delta p = p_f - p_i$ points along the final velocity here.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A ceiling fan starts from rest and speeds up uniformly to 900 rpm in 6.0 s. Its blades are 0.40 m long.

18) What is the angular acceleration of the fan?

Answer: (c) 15.7 rad/s²

$\omega = 900 \times 2\pi/60 = 94.2$ rad/s; $\alpha = \omega/t = 15.7$ rad/s².

19) How many revolutions does the fan make during these 6.0 s?

Answer: (c) 45 rev

$\theta = \frac{1}{2}\alpha t^2 = 283$ rad = 45 rev.

20) Compared with the blade tip, a point halfway along the blade has

Answer: (a) the same angular speed and half the linear speed.

Every point on a rigid body shares ω, and $v = \omega r$.

THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.

A 2.0 kg uniform rod 1.2 m long has a 0.50 kg point mass attached at each end.

21) What is the moment of inertia about an axis through the center of the rod, perpendicular to it?

Answer: (b) 0.6 kg·m²

$I = (1/12)(2)(1.2^2) + 2(0.5)(0.6^2) = 0.24 + 0.36 = 0.6$ kg·m².